Track A: Application

When Does Machine Learning Help? Predicting Pension Benefits under Defined Benefit and Defined Contribution Plans

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(a) Question

Machine learning is often expected to outperform standard econometric models when data are large and high-dimensional (Mullainathan and Spiess, 2017). Bajari et al. (2015), for example, show that random forests predict demand far more accurately than linear and logit models in a scanner dataset with over a million observations, when all models are given the same covariates. Kwon and Maliar (2026) bring gradient boosted trees to retirement and Social Security claiming in the HRS and find that the gains over the SSA's benchmark model are concentrated among defined contribution (DC) pension holders. They conjecture that DC arrangements are less transparent to the researcher than defined benefit (DB) pensions, whose benefits follow a formula, roughly $B = m \times YOS \times FAS$ (multiplier × years of service × final average salary). Building on both papers, this project applies a Bajari-style comparison of econometric and machine-learning models to test that conjecture directly, using pension benefit amounts as the outcome.

Question. Does the predictive gain of machine learning over linear benchmarks differ between DB and DC pension benefits?

Hypotheses. I expect machine learning to outperform OLS by a wider margin for DC than for DB. DB benefits are set by a plan formula, whereas DC benefits, apart from annual contribution limits, depend largely on individual choices about how much to contribute, how to invest and when to withdraw. If, contrary to this expectation, DB benefits also depend on many plan-specific factors that are not directly observed, machine learning may outperform OLS for both plan types.

Why it matters. Labor force participation at older ages depends heavily on expected retirement income, and pensions, as a steady stream of income, are central to how workers anticipate their living standards after retirement. Models of retirement decisions therefore need the benefit a worker expects before retiring, not the amount realized afterward (Stock and Wise, 1990). The HRS collects such expectations only partially: they are asked of DB participants, are often missing or noisy, and are essentially absent for DC plans, where only account balances are reported. A model trained on realized benefits can provide ex ante predictions as a function of the claiming age, which can then serve as an input for predicting retirement and later behavior. DB and DC plans determine benefits through different structures, so a single prediction method may not be equally accurate for both. This project identifies which approach is appropriate for each plan type.

(b) Data

Source. RAND HRS Longitudinal File 1992–2022 (V1), supplemented by the RAND HRS Fat Files for plan details not in the longitudinal file. Data cleaning follows Kwon and Maliar (2026): missing-value codes, harmonization of occupation and industry codes, and previous-wave values for earnings and assets.

Sample. Respondents born 1931–1953 (HRS, War Babies and Early Baby Boomer cohorts) who are observed working for the plan sponsor before claiming and who receive one employer pension in at least two consecutive waves. DB recipients hold a single DB plan on the job held in their last wave of work before claiming; DC recipients hold only DC plans on that job, with balances summed across plans. Plans from earlier jobs are not observed in the RAND file, but requiring a single pension in receipt excludes most respondents who draw benefits from more than one employer. Respondents with both plan types, combination plans or multiple DB plans on the same job, the self-employed, and military pensions are excluded.

Unit and size. The unit of observation is a pension recipient, and the outcome is the log real monthly pension averaged over the first two waves of receipt. Features are drawn from the work history before

claiming (earnings, job and plan characteristics, region) and from claiming conditions (claiming age relative to eligibility, marital status, Social Security receipt), with an extended set of household wealth and behavioral variables expected to matter mainly for DC. An initial count gives 955 DB and 432 DC recipients.

Access. Both files are public and require only free registration with the HRS. I already have the RAND longitudinal file and have constructed the sample; the Fat Files will be downloaded before the checkpoint.

(c) Method

All models predict the log real monthly benefit under squared-error loss and receive the same features, with identical transformations and missing-value treatment, so performance differs only in how each model maps features to benefits.

Model	Specification	What it captures
B0 (formula)	Log-linear form of the benefit rule: logs of $B = m \times YOS \times FAS$ give $\log B = \log m + \log YOS + \log FAS$, so DB benefits are regressed on log final average salary, log years of service and claiming-age dummies; DC benefits are regressed on log balance and age dummies, which proxy for the annuity factor.	How much the plan rule alone explains
B1 (OLS)	Linear regression on the full feature set	Linear, additive effects
B2 (LASSO)	Linear regression with an L1 penalty that sets some coefficients to zero	Linear effects, variable selection
RF (random forest)	Average of deep trees, each grown on a bootstrap sample with a random subset of features at each split (Breiman, 2001)	Nonlinearity, interactions; averaging reduces variance
GB (boosted trees)	Sum of shallow trees fitted sequentially, each to the errors left by the previous ones (Friedman, 2001)	Nonlinearity, interactions; sequential fitting reduces bias

Because the DC sample is small, performance is evaluated by nested cross-validation: an outer 5-fold split, repeated 10 times with fixed seeds, measures out-of-sample RMSE, and all hyperparameters are tuned by an inner 5-fold cross-validation within each training fold. The main measure is the relative gain, $(RMSE_OLS - RMSE_ML) / RMSE_OLS$, the proportional reduction in prediction error over OLS, computed separately for DB and DC; its spread across repetitions gives confidence intervals for the DB–DC difference. The comparison is repeated with the DB sample subsampled to the size of the DC sample.

A gain over OLS can come from using the same information more flexibly or from using more information. To separate the two, all models are estimated on both feature sets: the common set shows whether flexible functional forms help more for DC given the same information, and the extended set shows whether additional individual characteristics help DC more than DB.

(d) Feasibility

With about 1,400 observations and 30 features, all models (scikit-learn, XGBoost) run within minutes on a laptop CPU. The analysis sample and attrition table are already constructed and committed to the repository. By the week 11 checkpoint I expect to have the full feature set and B1, B2 and gradient boosted trees estimated for DB and DC on the common feature set, with nested cross-validation results. Fallbacks: (i) use only RAND longitudinal variables if merging the Fat Files is delayed; (ii) drop the extended feature set and focus on the common-set comparison; (iii) if DC results are too noisy, use the broader DC plan definition. SHAP decompositions, as in Kwon and Maliar (2026), will be used to relate predictions back to pension determinants; this part may be revised based on feedback.

I used an AI assistant to help structure the design and edit the text; all choices were reviewed by me.

References

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